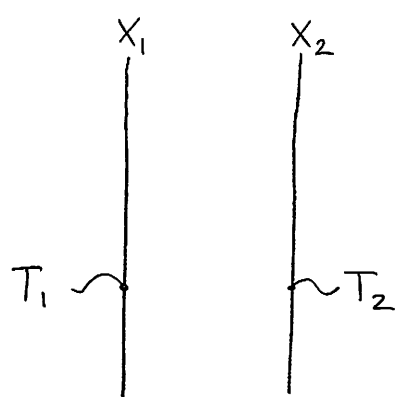


Wall with Variable Thermal Conductivity



1-D, S.S., no generation

When k weakly depends on T ,

$$k = a + bT \Rightarrow \text{units} \quad a \Rightarrow \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$b \Rightarrow \frac{\text{W}}{\text{m} \cdot \text{K}^2}$$

No need to use heat equation, even simpler

$$q_x = -k \frac{\partial T}{\partial x} \Rightarrow \text{Integrate}$$

$$\int_{x_1}^{x_2} q_x dx = - \int_{T_1}^{T_2} (a + bT) dT$$

We know that the heat flux is independent of location, (from the conservation of energy)

$$q_x(x_2 - x_1) = -a(T_2 - T_1) - \frac{b(T_2^2 - T_1^2)}{2} \Rightarrow \text{Factor}$$

$$q_x \Delta x = - \left(a + b \underbrace{\frac{T_2 + T_1}{2}}_{\bar{T}} \right) (T_2 - T_1)$$

$$\text{So: } q_x \Delta x = -(a + b\bar{T}) \Delta T$$

Rearranging, an interesting interpretation evolves

$$q_x = - \underbrace{(a + b\bar{T})}_{\bar{k}} \frac{\Delta T}{\Delta x}$$

$$\boxed{q_x = -\bar{k} \frac{\Delta T}{\Delta x}}$$

For the temperature distribution:

$$q_x = -k \frac{\partial T}{\partial x} = -(a + bT) \frac{\partial T}{\partial x}$$

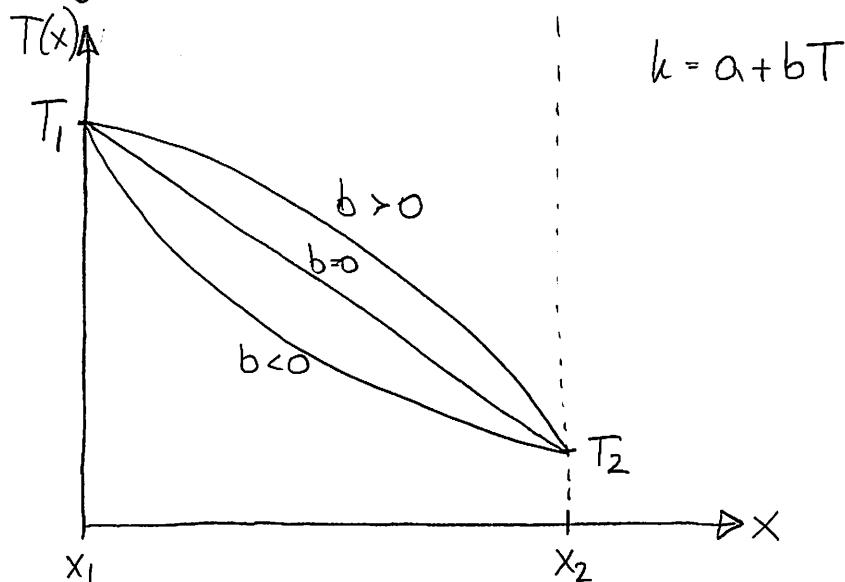
$$\int_{x_1}^x q_x dx = - \int_{T_1}^T (a + bT) dT \quad (\text{Note bounds of integration are different})$$

$$q_x(x - x_1) = -a(T(x) - T_1) - \frac{b(T(x)^2 - T_1^2)}{2} \Rightarrow \text{Quadratic equation.}$$

Solving our quadratic equation above:

$$T(x) = -\frac{a}{b} \pm \left\{ \left(\frac{a}{b} \right)^2 - \frac{2}{b} q_x (x - x_1) - T_1^2 - 2 \frac{a}{b} T_1 \right\}^{1/2}$$

Plotting our result:

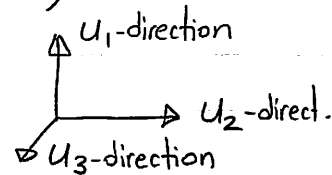


Very useful solution since many materials have a thermal conductivity temperature dependence that can be well approximated with a linear temperature dependent profile.

Conduction for General Shapes (An easier approach)

$\vec{V} = 0$ (no convection)

Steady State, $\dot{Q}''' = 0$, $k = \text{constant}$, 1-D



Define an orthogonal coordinate system U_1, U_2, U_3 (directions)

$$q'' = -k \frac{\partial T}{\partial s_1} \quad (\vec{q}'' = -k \vec{\nabla} T)$$

$$Q = A q'' = -k A(u_1) \frac{\partial T}{\partial s_1}$$

s_1, s_2, s_3 (lengths)

$$ds_1 = h_1 du_1$$

$$ds_2 = h_2 du_2$$

$$ds_3 = h_3 du_3$$

Rearranging and integrating (assuming $ds_1 = du_1$)

$$Q \frac{1}{k} \int_{(u_1)_a}^{(u_1)_b} \frac{du_1}{A(u_1)} = - \int_{T_a}^{T_b} dT = T_a - T_b = \Delta T$$

Note, we'll come to this later and reconcile.

$$Q = \frac{\Delta T}{\frac{1}{k} \int \frac{du_1}{A(u_1)}} = \frac{\Delta T}{R}$$

$$\boxed{R = \frac{1}{k} \int \frac{du_1}{A(u_1)}} \Rightarrow \begin{array}{c} T_1 \quad \xrightarrow[\quad Q \quad]{R} \quad T_2 \end{array}$$

Let's double check our previous solutions just to be sure:

① Slab; $u_1 = x$, $du_1 = dx$, $A(u_1) = \text{constant}$

$$R_{\text{slab}} = \frac{1}{kA} \int du_1 = \frac{1}{kA} \int dx = \frac{L}{kA}$$

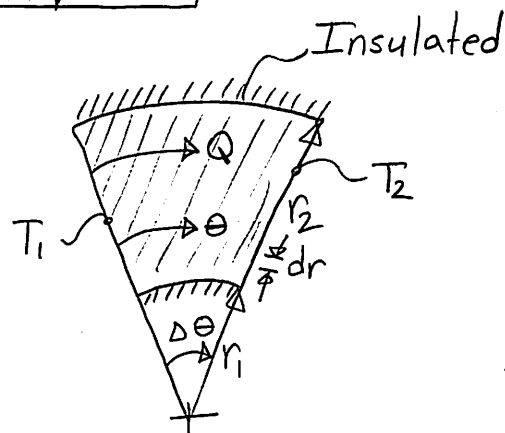
② Cylinder; $u_1 = r$, $du_1 = dr$, $A(u_1) = A(r) = 2\pi r l$

$$R_{\text{cyl}} = \frac{1}{k} \int_{r_1}^{r_2} \frac{dr}{2\pi r l} = \frac{1}{2\pi k l} \int_{r_1}^{r_2} \frac{dr}{r} = \frac{\ln(r_2/r_1)}{2\pi k l}$$

③ Sphere; $u_1 = r$, $du_1 = dr$, $A(u_1) = A(r) = 4\pi r^2$

$$R_{\text{sph}} = \frac{1}{k} \int_{r_1}^{r_2} \frac{dr}{4\pi r^2} = \frac{1}{4\pi k} \int_{r_1}^{r_2} \frac{dr}{r^2} = \frac{1}{4\pi k} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Example #4



$T = T(\theta)$, l = thickness into page

Note, more complex since heat flux is also a function of r since heat flow through the region closer to the center happens easier than the outside (Lower length and thermal resistance)

Using our general formulation: $\left. \begin{array}{l} u_1 = \theta, \quad ds_1 = r d\theta \\ d(A(u_1)) = l dr \end{array} \right\} \text{For radial Coordinates}$

$$dQ = -k d(A(u_1)) \frac{\partial T}{\partial s_1} = -k d(A(u_1)) \frac{1}{r} \cdot \frac{\partial T}{\partial \theta}$$

$$= -k l dr \frac{\partial T}{r \partial \theta} = -k l \cdot \frac{1}{r} dr \frac{\partial T}{\partial \theta}$$

$$\int dQ = \int -k l \frac{\partial T}{\partial \theta} \cdot \frac{dr}{r}$$

$$Q = -k l \frac{\partial T}{\partial \theta} \int_{r_1}^{r_2} \frac{dr}{r} = -k l \frac{\partial T}{\partial \theta} \cdot \ln(r_2/r_1)$$

$$Q = -k l \ln(r_2/r_1) \frac{\partial T}{\partial \theta} = \underbrace{\frac{k l \ln(r_2/r_1)}{\Delta \theta}}_{R^{-1}} \cdot \Delta T$$

$$\boxed{R = \frac{\Delta \theta}{k l \ln(r_2/r_1)}} \Rightarrow \text{Thermal resistance of the arc}$$

Let's check the bounds of our solution to see if it makes sense: What if $r_1 \approx r_2 = r_2 - \delta$

$$\ln\left(\frac{r_2}{r_1}\right) = \ln\left(\frac{r_2}{r_2 - \delta}\right) = \ln\left(\frac{1}{1 - \frac{\delta}{r_2}}\right) = \ln\left[\frac{1 + \frac{\delta}{r_2}}{(1 - \frac{\delta}{r_2})(1 + \frac{\delta}{r_2})}\right]$$

$$= \ln \left[\frac{1 + \frac{\delta}{r_2}}{1 + \cancel{\frac{\delta}{r_2}} - \cancel{\frac{\delta}{r_2}} - \frac{\delta^2}{r_2^2}} \right], \text{ since } \frac{\delta}{r_2} < 1$$

$$= \ln \left(1 + \frac{\delta}{r_2} \right) \quad \frac{\delta^2}{r_2^2} \ll 1, \approx 0$$

Using series expansion:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots \frac{(-x)^{N+1}}{N+1} + \dots$$

$$x = \frac{\delta}{r_2}$$

$$\ln \left(1 + \frac{\delta}{r_2} \right) = \frac{\delta}{r_2} - \cancel{\frac{\delta^2}{2r_2^2}} + \cancel{\frac{\delta^3}{3r_2^3}} + \dots$$

$$\ln \left(1 + \frac{\delta}{r_2} \right) \approx \frac{\delta}{r_2}$$

So plugging back into our resistance calculation:

$$R = \frac{\Delta \theta}{k l \ln(r_2/r_1)} = \frac{\Delta \theta r_2}{\underbrace{k l \delta}_{\frac{L}{kA}}} \Rightarrow \begin{array}{l} l \delta = A \text{ (Area)} \\ \Delta \theta r_2 = L \text{ (Length)} \end{array}$$

$\frac{L}{kA} \Rightarrow \text{Works like a charm!}$

General Orthogonal Coordinate System (Pure Conduction)
Defining u_1, u_2, u_3 (3 coordinate dimensions)

$$\left. \begin{array}{l} ds_1 = h_1 du_1 \\ ds_2 = h_2 du_2 \\ ds_3 = h_3 du_3 \end{array} \right\} \text{For example} \Rightarrow \text{Cartesian} \quad \begin{array}{l} h_1 = 1, du_1 = dx \\ h_2 = 1, du_2 = dy \\ h_3 = 1, du_3 = dz \end{array}$$

We can generalize the following:

$$\boxed{q'' = -k \frac{1}{h_1} \cdot \frac{\partial T}{\partial u_1}}$$

$$Q = \iint q'' ds_2 ds_3 = \iint q'' h_2 h_3 du_2 du_3$$

$$Q = -k \frac{2T}{2u_1} \iint \frac{h_2 h_3}{h_1} du_2 du_3$$

$$R = \frac{1}{k} \int_{(u_1)_a}^{(u_1)_b} \frac{du_1}{\iint \frac{h_2 h_3}{h_1} du_2 du_3}$$

$\Rightarrow$ The most usefull & general form you'll learn. Very powerfull.

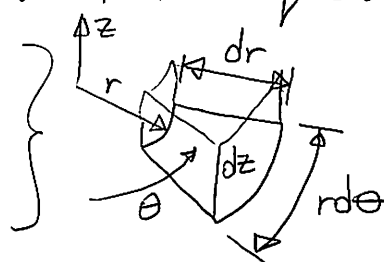
Let's check with two quick examples $\Rightarrow$ radial & spherical

① Radial:

$$u_1 = r$$

$$u_2 = \theta$$

$$u_3 = z$$



$$ds_1 = (1) dr \Rightarrow h_1 = 1$$

$$ds_2 = r d\theta \Rightarrow h_2 = r$$

$$ds_3 = (1) dz \Rightarrow h_3 = 1$$

Back substituting into our formulation

$$Q = -k \frac{2T}{2r} \int_0^L \int_0^{2\pi} \frac{r(\cancel{r})}{(\cancel{r})} d\theta dz = -k \frac{2T}{2r} \int_0^L dz \int_0^{2\pi} r d\theta$$

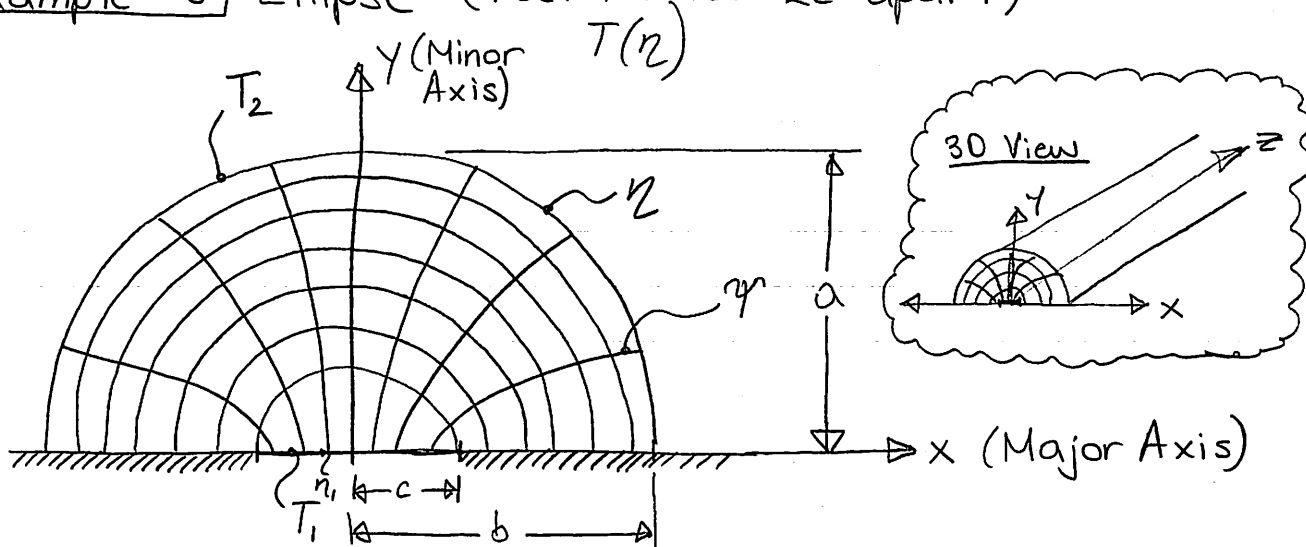
$$= -k \underbrace{(2\pi r)}_A L \frac{2T}{2r} = -k A \frac{2T}{2r}$$

$$R = \frac{1}{k} \int_{r_1}^{r_2} \frac{dr}{\int_0^L \int_0^{2\pi} \frac{r(\cancel{r})}{(\cancel{r})} d\theta dz}$$

$$= \frac{1}{k} \int_{r_1}^{r_2} \frac{dr}{r \int_0^L dz \int_0^{2\pi} d\theta} = \frac{1}{2\pi k L} \int_{r_1}^{r_2} \frac{dr}{r} = \frac{\ln(r_2/r_1)}{2\pi k L} = R_{cyl}$$

We see that the formalism works. Try it for spherical.

Example #5 | Ellipse (Foci located $2c$ apart)



$$\begin{aligned} U_1 &= \eta & h_1 &= \text{unknown yet (we will solve for it)} \\ U_2 &= \psi & h_2 &= \text{unknown yet} \\ U_3 &= z, & h_3 &= 1 \end{aligned}$$

First we need to solve for h_1 & h_2 . We can use some nice rules & identities for ellipses:

Rule #1:

Coordinate Transformation:

$$b^2 - a^2 = c^2 \Rightarrow \begin{aligned} x &= c \cosh \eta \cos \psi & b &= c \cosh \eta_1 \\ y &= c \cosh \eta \sin \psi & a &= c \sinh \eta_1 \end{aligned}$$

$$(b^2 - a^2) = c^2 (\cosh^2 \eta_1 - \sinh^2 \eta_1)$$

$$1 \Rightarrow \text{so } b^2 - a^2 = c^2 \text{ (we are OK!)}$$

Some identities we'll need are

$$\cosh \eta = \frac{e^\eta + e^{-\eta}}{2}$$

$$\sinh \eta = \frac{e^\eta - e^{-\eta}}{2}$$

$$\vec{r} = x\vec{i} + y\vec{j} \Rightarrow \left| \frac{d\vec{r}}{d\eta} \right| = h_1 = c (\sinh^2 \eta \cos^2 \psi + \cosh^2 \eta \sin^2 \psi)^{1/2}$$

$$\text{(Coordinate Transformation)} \quad \left| \frac{d\vec{r}}{d\psi} \right| = h_2 = c (\cosh^2 \eta \sin^2 \psi + \sinh^2 \eta \cos^2 \psi)^{1/2} \quad \left. \vphantom{\left| \frac{d\vec{r}}{d\psi} \right|} \right\} h_1 = h_2$$

This makes our lives a lot easier:

$$R = \frac{1}{h} \int_0^{n_1} \frac{dn}{\int_0^\pi \int_0^L d\varphi dz} = \frac{n_1}{\pi k L}$$

We know that: $\frac{a+b}{c} = \cosh n_1 + \sinh n_1$

$$= \frac{e^{n_1} + e^{-n_1}}{2} + \frac{e^{n_1} - e^{-n_1}}{2} = e^{n_1}$$

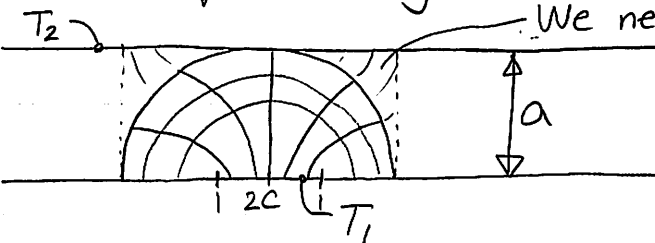
$$n_1 = \ln\left(\frac{a+b}{c}\right)$$

So our solution becomes:

$$Q = \frac{\Delta T \pi k L}{\ln\left(\frac{a+b}{c}\right)} \quad R = \frac{\ln\left(\frac{a+b}{c}\right)}{\pi k L}$$

Note, this solution is extremely usefull since it can be used to solve the following cases:

Case 1: Spot Welding

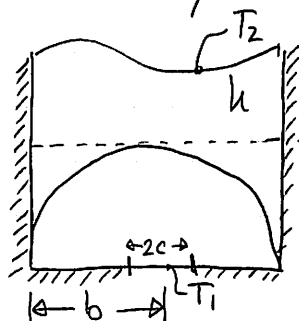


We neglect this resistance. But because spreading resistance here is high, safe approximation.

$$b^2 = \sqrt{a^2 + c^2}$$

Using the above resistance puts an upper bound on our heat transfer Q by assuming cylindrical spreading.

Case 2:

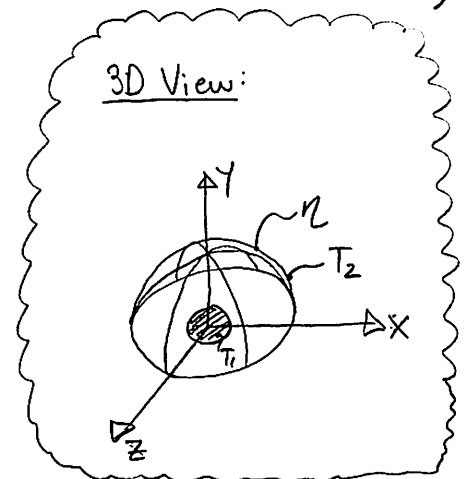
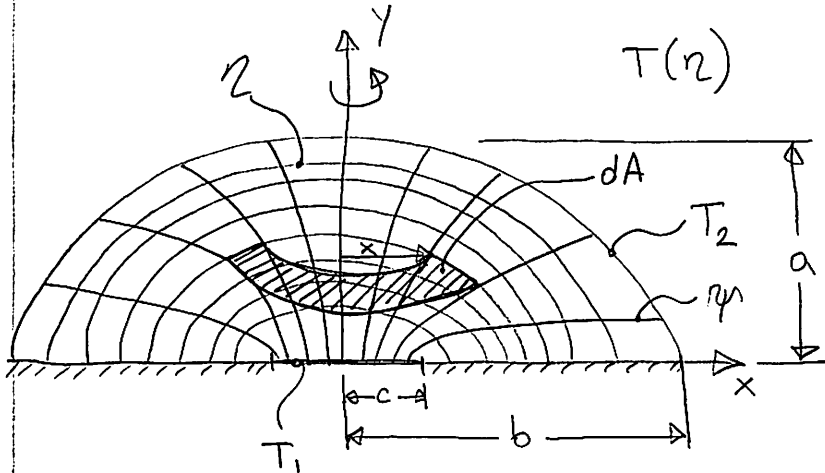


$$\Rightarrow R_1 = \frac{L}{kA}$$

$$\Rightarrow R_c = \frac{\ln\left(\frac{a+b}{c}\right)}{\pi k L} \Rightarrow a = \sqrt{b^2 - c^2}$$

Broken down into 2 resistances.
Good approximation.

Revolved Ellipsoid b (Major axis), a (Minor axis), $2c$ (Distance between foci)



$$q'' = -k \frac{1}{h_1} \cdot \frac{\partial T}{\partial \eta}$$

$$dA = 2\pi x \cdot h_2 \cdot d\psi$$

$$Q = \iint q'' dA = 2\pi k \frac{\partial T}{\partial \eta} \iint \frac{x h_2}{h_1} d\psi$$

$$Q = -2\pi k c \cosh \eta \frac{\partial T}{\partial \eta} \int_0^{\pi/2} \cos \psi d\psi$$

$$= -2\pi k c \cosh \eta \frac{\partial T}{\partial \eta}$$

$$\Delta T = Q \underbrace{\frac{1}{2\pi k c} \int_0^{\eta_1} \frac{d\eta}{\cosh \eta}}_R$$

$$R = \frac{1}{2\pi k c} \int_0^{\eta_1} \frac{2 d\eta}{e^\eta + e^{-\eta}} = \frac{1}{\pi k c} \int_0^{\eta_1} \frac{e^\eta d\eta}{e^{2\eta} + 1}, \text{ let } \lambda = e^\eta$$

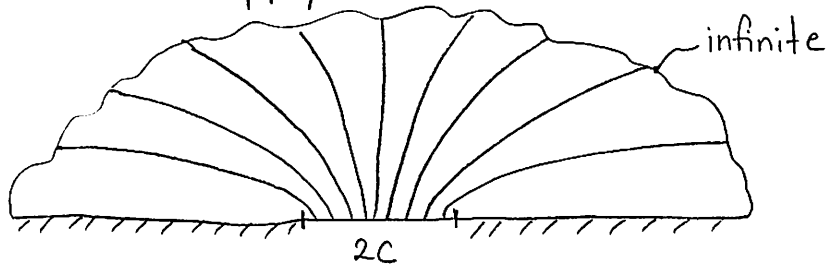
$$R = \frac{1}{\pi k c} \int_0^{e^{\eta_1}} \frac{d\lambda}{\lambda^2 + 1}$$

$$R = \frac{1}{\pi k c} \left[\tan^{-1}(e^{\eta_1}) - \frac{\pi}{4} \right] \Rightarrow \text{we know that } \eta_1 = \ln\left(\frac{a+b}{c}\right)$$

$$e^{\eta_1} = \frac{a+b}{c}$$

$$\boxed{R = \frac{1}{\pi k c} \left[\tan^{-1}\left(\frac{a+b}{c}\right) - \frac{\pi}{4} \right]}$$

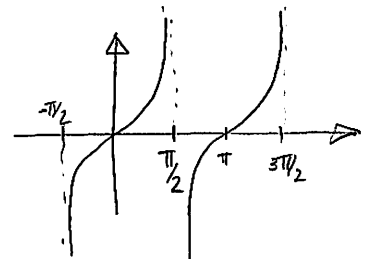
So if we apply our solution to a semi-infinite body:



$$a \rightarrow \infty$$

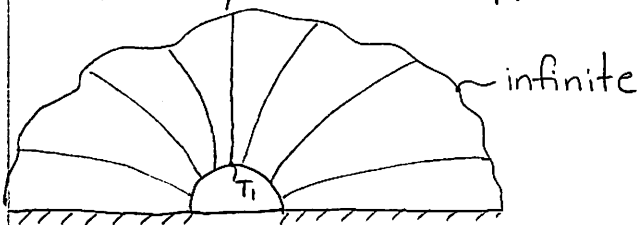
$$b \rightarrow \infty$$

$$R = \frac{1}{\pi k c} \left[\underbrace{\tan^{-1} \left(\frac{a+b}{c} \right)}_{\tan^{-1}(\infty) = \frac{\pi}{2}} - \frac{\pi}{4} \right]$$



$$R = \frac{1}{\pi k c} \left[\frac{\pi}{2} - \frac{\pi}{4} \right] = \frac{1}{4 k c}$$

When NASA was working with this problem in the 50's and 60's, they used an approximation using spherical resistance.



$$R_{\text{approx}} \approx \frac{1}{2\pi k} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

note, divide by 2
not 4 since
half-sphere.

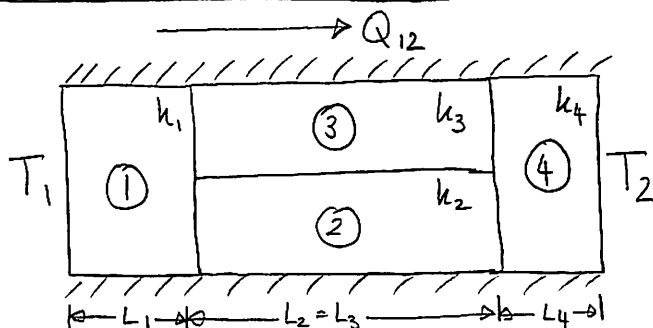
$r_2 \rightarrow \infty$
 $r_1 \rightarrow c$

$$R_{\text{approx}} = \frac{1}{2\pi k c} \quad (\approx 50\% \text{ off})$$

$$R_{\text{REAL}} = \frac{1}{4 k c} < R_{\text{approx}}$$

Composite Wall Problems

END OF LECTURE 5



There are two ways to analyze this problem (both giving different answers)